Mahmoud Alyosify

Computer Vision Engineer | Self-Supervised Representation Learning & Visual Retrieval
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PROFESSIONAL SUMMARY

Computer Vision Engineer focused on **representation learning, self-supervised pre-training and visual retrieval**. Built a full **SimCLR** pipeline with a **42-experiment augmentation sweep** and a semi-supervised **SupCon** extension reaching **84.30% top-1 accuracy**, then shipped it as a deployable **ONNX + FAISS** retrieval service. Earlier work covers **video-based physiological signal extraction (rPPG)** — a contactless vital-sign system graded **A+** and taken to **Microsoft Imagine Cup 2023 semi-finals** — and **CNN classification over time–frequency spectrograms** for deepfake audio detection. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, while working as an **AI Engineer Intern at Injaz Tech**. Strong on the part most vision demos skip: rigorous augmentation ablation, honest evaluation and deployment.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Deep Learning, Generative AI, Large Language Models, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation from video via **rPPG** and wavelet filtering. **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Computer Vision	CNNs, Image Classification, Self-Supervised Learning, Contrastive Learning (SimCLR, SupCon), Representation & Embedding Learning, Image Retrieval, Data Augmentation Design, Transfer Learning, Video-Based Signal Extraction (rPPG), Spectrogram-Based Classification
Frameworks & Tools	PyTorch, TensorFlow, OpenCV, Scikit-learn, NumPy, Hugging Face, Librosa, Optuna, SHAP
Deployment	ONNX, FAISS Vector Search, Docker, FastAPI, Streamlit, AWS, Latency Profiling, Git/GitHub, Linux
Evaluation	Top-1 / Top-k Accuracy, Retrieval Metrics, Linear Probing, Ablation Studies, Augmentation Sweeps, Reproducible Experiment Design
Also	Large Language Models, Fine-Tuning (LoRA, QLoRA), RAG (LangChain, LangGraph), Multimodal & Agentic Systems, Reinforcement Learning (PPO, DQN)
Programming	Python, C++, C, SQL, C#, MATLAB, Bash

SELECTED COMPUTER VISION PROJECTS

SimCLR-Vision-SSL — Self-Supervised Contrastive Representation Learning — *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Implemented a complete **SimCLR** pipeline — augmentation stack, projection head, NT-Xent contrastive objective and downstream evaluation — trained without labels.
- Ran a **42-experiment augmentation sweep** to isolate which transformations and combinations actually drive representation quality, rather than accepting published defaults.
- Extended the pipeline with a semi-supervised **SupCon** objective, reaching **84.30% top-1 accuracy** and evaluating learned embeddings on both classification and **image retrieval**.
- Exported the encoder to **ONNX** and served embeddings through a **FAISS** index for deployment-ready nearest-neighbour visual search.

Vitalism Solution — Contactless Vital-Sign Estimation from Video — *2023*
Computer Vision · rPPG · Wavelet Filtering · Signal Processing

- Built an AI system estimating vital signs in real time from ordinary video by extracting **remote photoplethysmography (rPPG)** signals from facial regions and denoising them with wavelet filtering — no contact sensors.
- Graded **A+** as the undergraduate capstone and selected as a **Semi-Finalist at Microsoft Imagine Cup 2023**.

AudioShield — CNN Deepfake Detection over Mel-Spectrograms — *2023*
TensorFlow · CNNs · Librosa · Streamlit

- Trained a **CNN** to classify authentic versus synthetic audio from **mel-spectrogram** images, applying vision-style convolutional modelling to time–frequency representations.
- Deployed as a Streamlit application with real-time prediction and automated PDF reporting, targeting fraud detection, digital forensics and media authentication.

Automated Multimodal Agent — Document Understanding to Narrated Presentation *2026*
Multimodal AI · Document Parsing · Generative AI

- Built an autonomous pipeline that parses visual and textual document structure and regenerates it as a fully narrated presentation, coordinating extraction, layout generation and audio synthesis.

Additional Projects

- **HORUS Sentinel** — multi-agent intelligence platform correlating multi-source findings into a knowledge graph with a self-hosted fine-tuned LLM.
- **SCRAP** — supervised risk-prediction pipeline forecasting satellite collision risk **48+ hours** ahead.
- **Also:** MinimalLM (MSc research) · Horus-OSINT (Llama-3-8B fine-tuning on AWS) · Diabetes Onset Prediction.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Built an **agentic RAG** assistant (LangChain, LangGraph, FAISS, re-ranking) served via **FastAPI** with Docker and 136 tests; top-1 retrieval accuracy **75% to 94%**.

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

Machine Learning Specialization — DeepLearning.AI

HCIP – AI (Huawei)

Semi-Finalist — Microsoft Imagine Cup 2023

Digilians Presidential Scholarship — Queen's University

AWS Academy — ML Foundations, ML for NLP, Data Engineering

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.